Kwame Nkrumah University of
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ME 281 – Engineering Materials

Lecture 5: Mechanical Properties of Metals

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Lesson Objectives

- Stress and Strain
  - ✓ Tension
  - ✓ Compression
  - ✓ Shear
  - ✓ Torsion
- Elastic Behaviour
- Plastic Behaviour
  - ✓ Yield Strength
  - ✓ Tensile Strength
  - ✓ Ductility
  - ✓ Toughness
  - ✓ Hardness
- Toughness and Ductility

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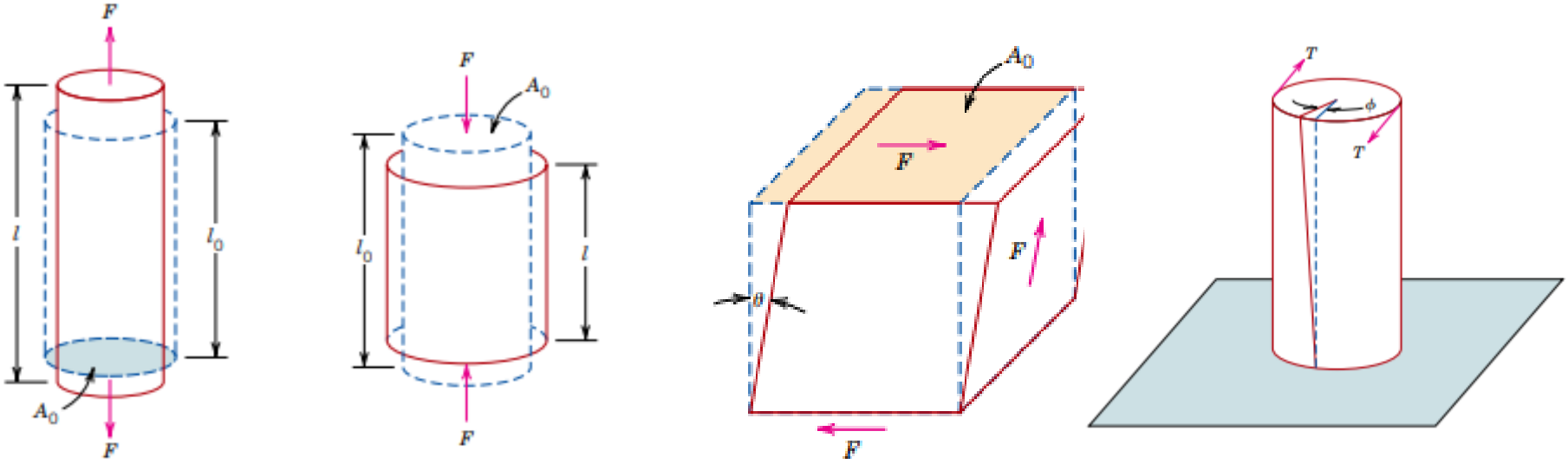
Introduction

- Most engineering materials are subjected to loads in practise.
- The mechanical behaviour of a material describes its response or deformation to an applied load or force.
- Key mechanical design properties are **stiffness, strength, hardness, ductility,** and **toughness.**
- The mechanical properties of materials are ascertained by performing carefully designed laboratory experiments that replicate as nearly as possible the service conditions.

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Concepts of Stress and Strain

- The mechanical behaviour of a material subjected to a uniformly applied static load or load that changes relatively slowly with time can be determined by a simple stress–strain test.
- A load may be applied in tension, compression, or shear
- In engineering practice many loads are torsional rather than pure shear.



Tension **Compression** **Shear** **Torsion**

Tension Test

Procedure

- A material is gripped at both ends by an apparatus, which slowly pulls lengthwise on the piece until it fractures.
- The pulling force is called a load, which is plotted against the material length change, or displacement.
- The load is converted to a stress value and the displacement is converted to a strain value.

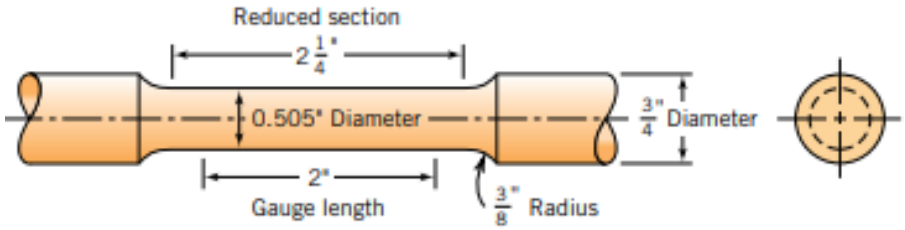


Fig. A standard tensile specimen with circular cross section

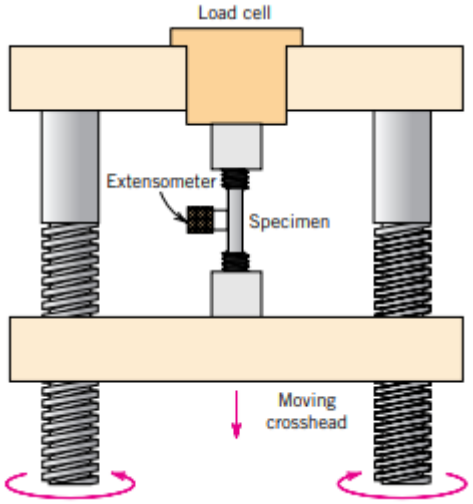


Fig. Tensile stress-strain Test setup

Shear and Strain (Tension and Compression)

- To compare specimens of different sizes, the load is calculated per unit area.

Engineering stress is given by

$$\sigma = F / A_o$$

Where

F is load applied perpendicular to specimen's cross-section;

$\mathbf{A_o}$ is cross-sectional area before application of the load.

Engineering strain is given by

$$\varepsilon = \Delta l / l_o \ (\times 100\ \%)$$

Where

$\boldsymbol{\Delta l}$ is change in length and $\boldsymbol{l_o}$ is the original length.

- These definitions of stress and strain allow one to compare test results for specimens of different cross-sectional area $\mathbf{A_o}$ and of different length $\boldsymbol{l_o}$.
- Stress and strain are positive for tensile loads, negative for compressive loads

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Shear and Strain (Shear and Torsion)

- **Shear stress** is given by

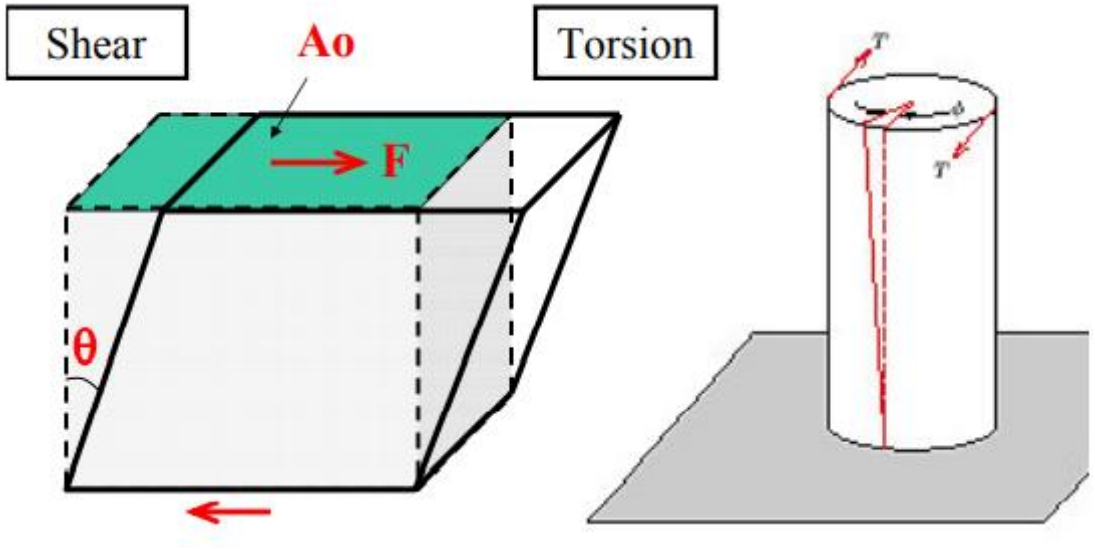
$$\tau = \frac{F}{A_o}$$

Where **F** is load applied parallel to the upper and lower faces each of which has an area A_o.

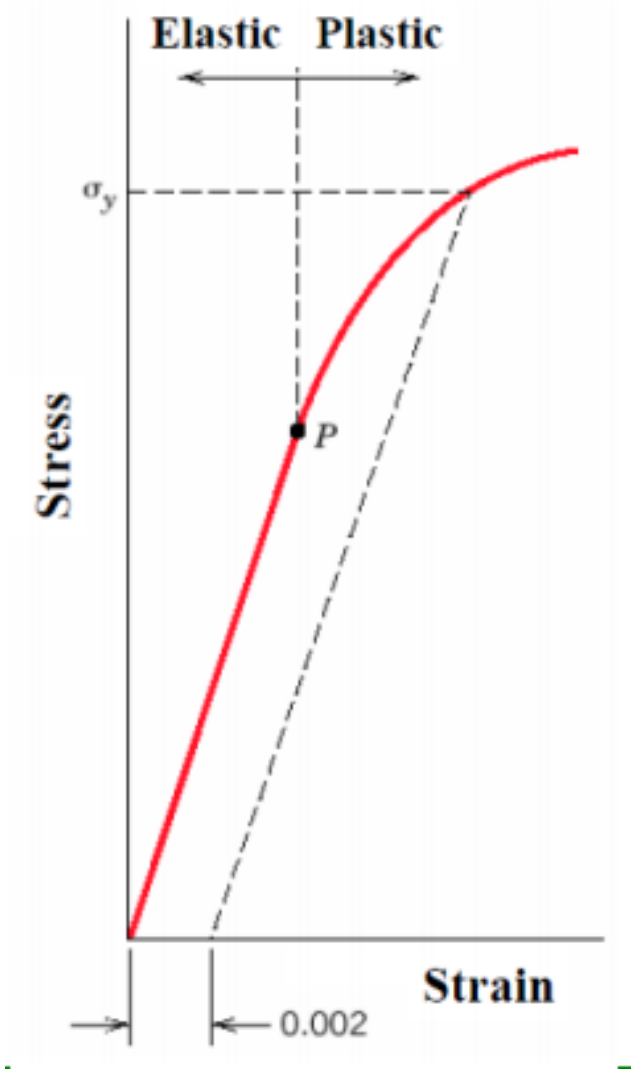
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Shear and Strain (Shear and Torsion)

- **Torsion** is variation of pure shear.
- The shear stress in this case is a function of applied torque T,
- **Shear strain** is related to the angle of twist, ∅.



Stress-Strain Behaviour



Elastic deformation

- Reversible: when the stress is removed, the material returns to the dimensions it had before the loading.
- Usually strains are small (except for the case of some plastics, e.g. rubber).

Plastic deformation

- Irreversible: when the stress is removed, the material does not return to its original dimensions.

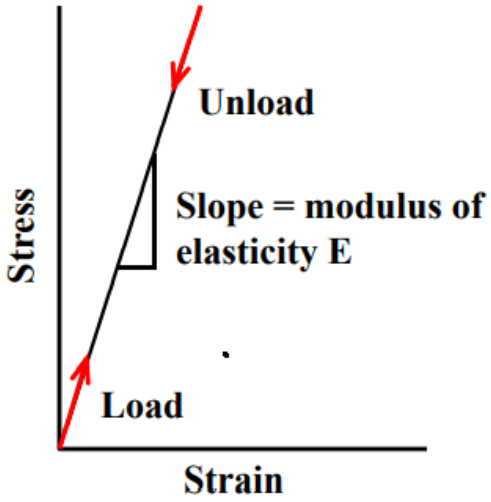
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Stress-Strain Behaviour – Elastic Deformation

- In tensile tests, if the deformation is elastic, the stress-strain relationship is called Hooke's law and is given by

$$\sigma = E\,\varepsilon$$

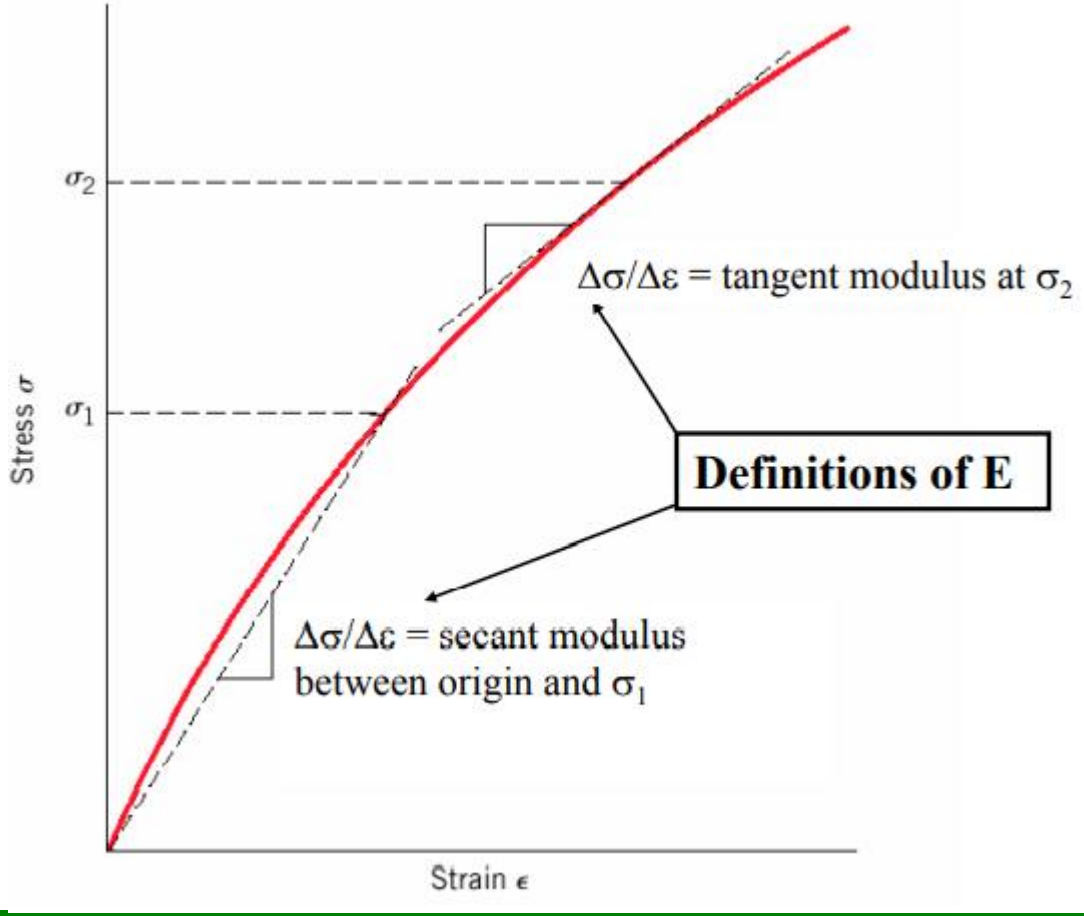
- **E** is Young's modulus or modulus of elasticity, and has the same units as σ, N/m^2 or Pa.
- Higher **E** → higher "stiffness"
- Deformation in which stress and strain are proportional is called elastic deformation



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Elastic Deformation: Nonlinear Elastic Behaviour

- Elastic deformation is non linear but still reversible in some materials such as polymers and concrete.



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Elastic Deformation

- For elastic deformation, the relationship between shear stress and shear strain are related by

$$\tau = G\gamma$$

where **G** is the shear modulus.

- The shear modulus is the slope of the linear elastic region of the shear stress–strain curve

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Elastic Deformation: Anelasticity

- So far we have assumed that elastic deformation is time independent (i.e. applied stress produces instantaneous elastic strain)
- However, in reality elastic deformation takes time - continues after initial loading, and after load release.
- This time dependent elastic behaviour is known as **anelasticity.**
- The effect is normally small for metals but can be significant for polymers ("visco-elastic behavior").

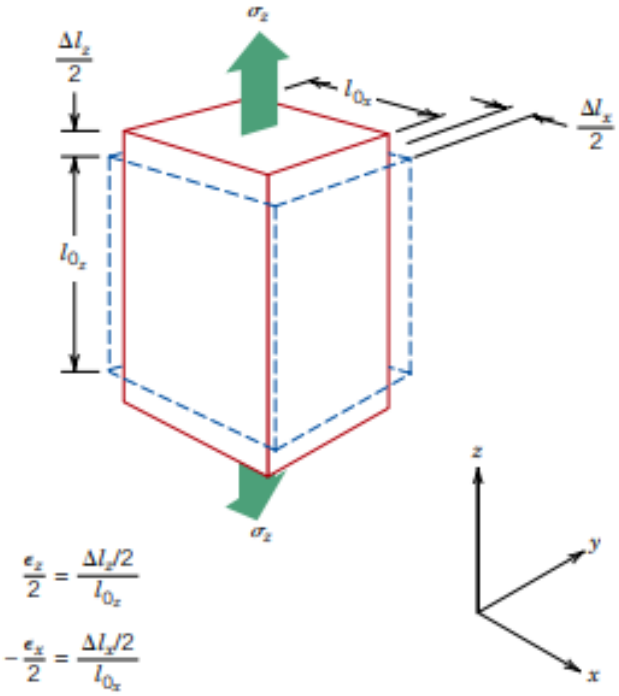
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Elastic Deformation – Poisson's Ratio

- Materials subject to tension shrink laterally. Those subject to compression, bulge.
- The ratio of lateral and axial strains is called the **Poisson's ratio υ**.

$$\nu = -\frac{\varepsilon_x}{\varepsilon_z} = -\frac{\varepsilon_y}{\varepsilon_z}$$

- Sign in the above equations shows that lateral strain is in opposite sense to longitudinal strain
- Maximum value of ν is 0.50.
- For isotropic materials **E = 2G(1+υ)**



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Plastic Deformation (Stress-strain Behaviour)

- For most metallic materials, elastic deformation persists only to strains of about 0.005.
- As the material is deformed beyond this point, the stress is no longer proportional to strain, and permanent, non-recoverable, or **plastic deformation** occurs.
- The transition from elastic to plastic is a gradual process for most metals.
- Some curvature results at the onset of plastic deformation, which increases more rapidly with rising stress.

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Plastic Deformation (Stress-strain Behaviour)

- From an atomic perspective, plastic deformation corresponds to the breaking of bonds with original atom neighbours and then re-forming bonds with new neighbours as large numbers of atoms or molecules move relative to one another; upon removal of the stress they do not return to their original positions.

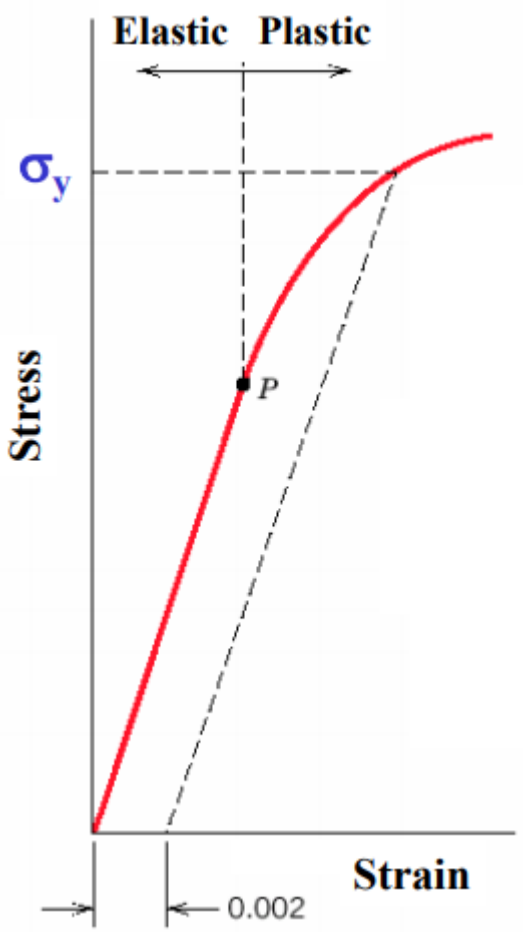
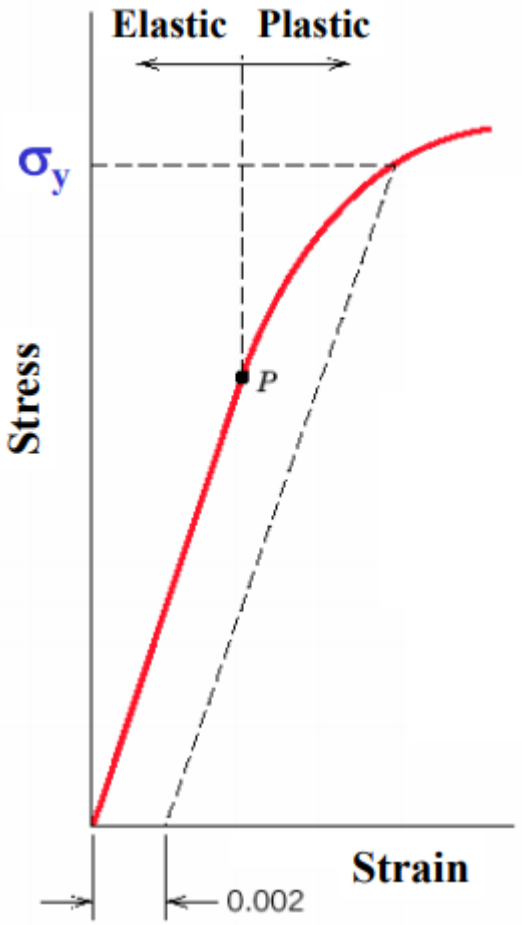


Fig. Stress-strain behaviour for a metal showing elastic and plastic deformation

Tensile Properties – Yielding and Yield Strength

- **Yielding** occurs at the stress level where plastic deformation begins.
- The yielding point can be determined as the initial departure from linearity of the stress–strain curve, called the **proportional limit.**
- **Yield strength σ_y** is chosen as that causing a permanent strain of 0.002.
- **Yield point P** – The point at which the strain deviates from being proportional to the stress (the proportional limit).
- The **Yield stress** is a measure of resistance to plastic deformation.



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Tensile Properties – Yielding and Yield Strength

- Some steels and other metals (low carbon steels) have the elastic–plastic transition occurring abruptly in a **yield point phenomenon.**
- At the upper yield point, plastic deformation is initiated with an apparent decrease in engineering stress.
- Continued deformation fluctuates slightly about some constant stress value, termed the lower yield point; stress subsequently rises with increasing strain.
- For such metals that display this effect, the yield strength is taken as the average stress that is associated with the lower yield point.
- The magnitude of the yield strength for a metal is a measure of its resistance to plastic deformation.

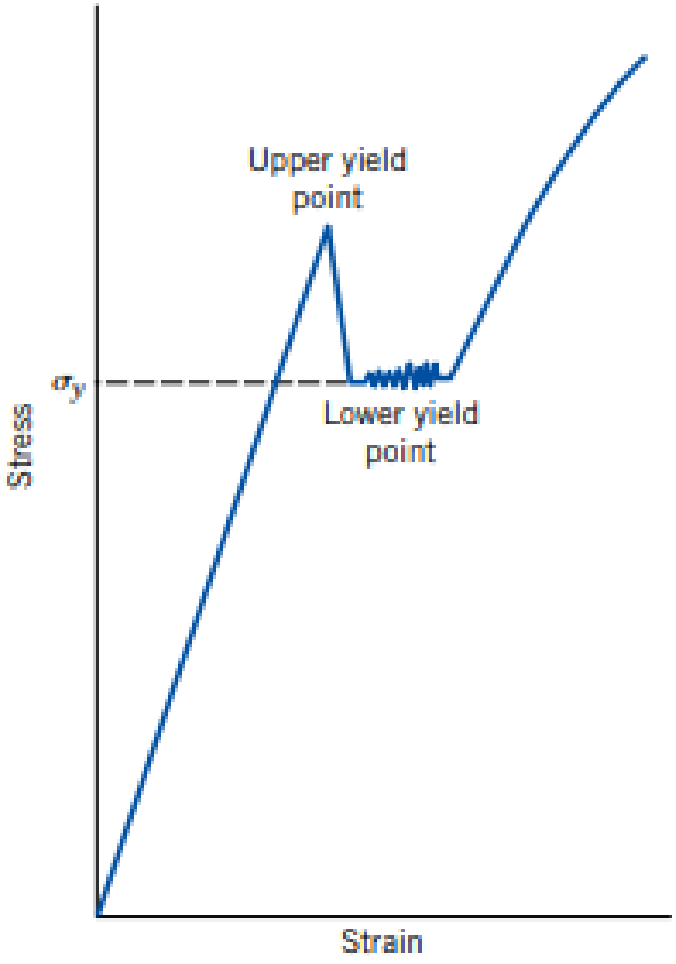
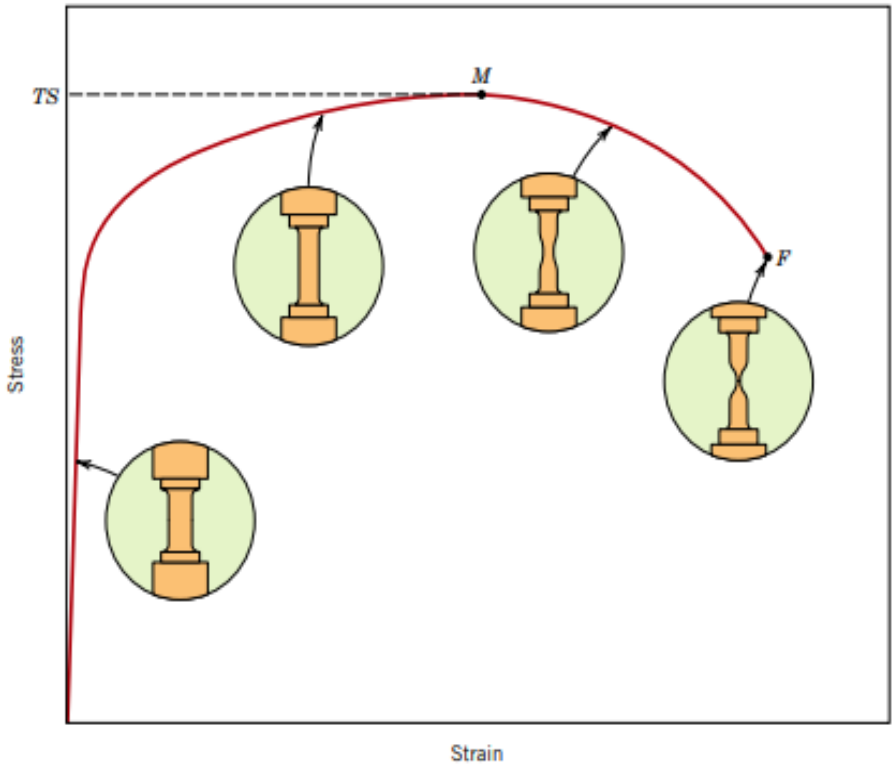


Fig. Stress–strain behaviour showing the yield point phenomenon

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Tensile Strength



- After yielding, the stress necessary to continue plastic deformation in metals increases to a maximum, point M and then decreases to the eventual fracture, point F.
- The **tensile strength** is the stress at the maximum on the engineering stress–strain curve and it corresponds to the maximum stress that can be sustained by a structure in tension.
- If the tensile strength (stress) is applied and maintained, fracture will result.
- **Necking** begins at the tensile strength and fracture ultimately occurs at the neck.

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Ductility

- Ductility is a measure of the deformation at fracture
- Ductility is defined by percent elongation as

$$\%EL = \left(\frac{l_f - l_0}{l_0} \right) \times 100$$

- It is also defined by percent reduction in area as

$$\%RA = \left(\frac{A_0 - A_f}{A_0} \right) \times 100$$

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Example: Mechanical Property Determinations from Stress–Strain Plot

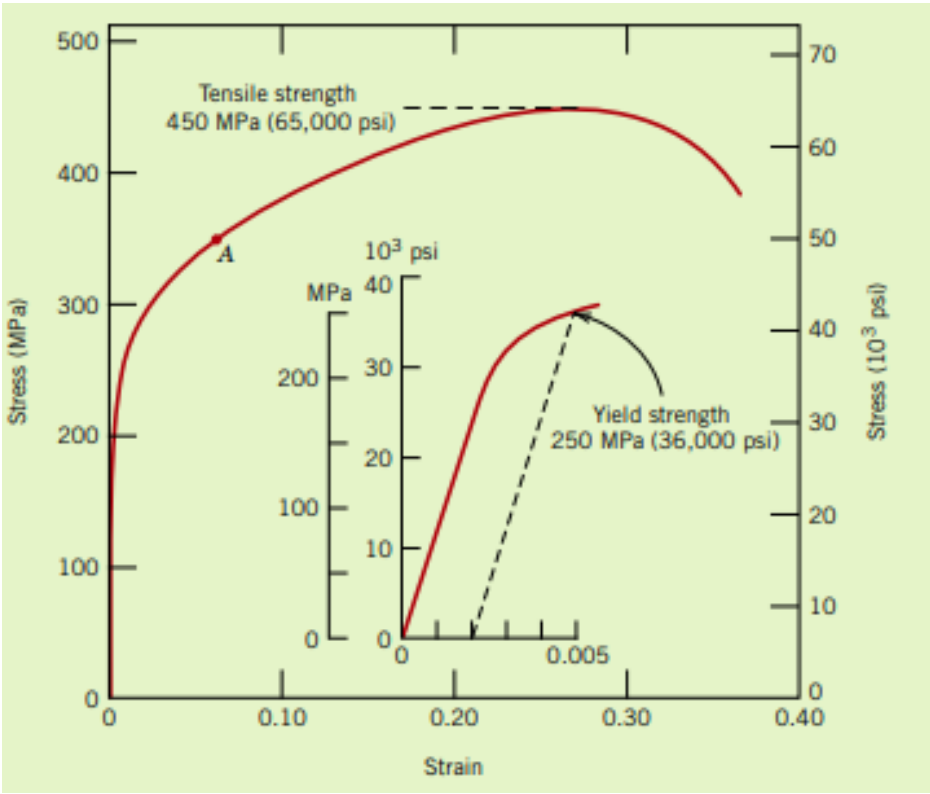
From the tensile stress–strain behaviour for the brass specimen shown in the Figure attached, determine the following:

(a) The modulus of elasticity

(b) The yield strength at a strain offset of 0.002

(c) The maximum load that can be sustained by a cylindrical specimen having an original diameter of 12.8 mm

(d) The change in length of a specimen originally 250 mm long that is subjected to a tensile stress of 345 MPa (50,000 psi)



Solution

(a) The modulus of elasticity is the slope of the elastic or initial linear portion of the stress–strain curve. The slope of this linear region the change in stress divided by the corresponding change in strain:

$$E = slope = \frac{\Delta\sigma}{\Delta\varepsilon} = \frac{\sigma_2 - \sigma_1}{\varepsilon_2 - \varepsilon_1}$$

$$E = \frac{(150 - 0)MPa}{(0.0016 - 0)} = 93.8GPa$$

(b) The 0.002 strain offset line is constructed as shown in the inset; its intersection with the stress–strain curve is at approximately 250 MPa (36,000 psi), which is the yield strength of the brass.

(c) The maximum load that can be sustained by the specimen is taken to be the tensile strength, from Figure, 450 MPa (65,000 psi). Solving for F, the maximum load, yields

$$F = \sigma A_o = \sigma\pi\left(\frac{d_o}{2}\right)^2 = (450MPa) \times \pi\left(\frac{12.8 \times 10^{-3}m}{2}\right)^2 = 57900N$$

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Solution cont.

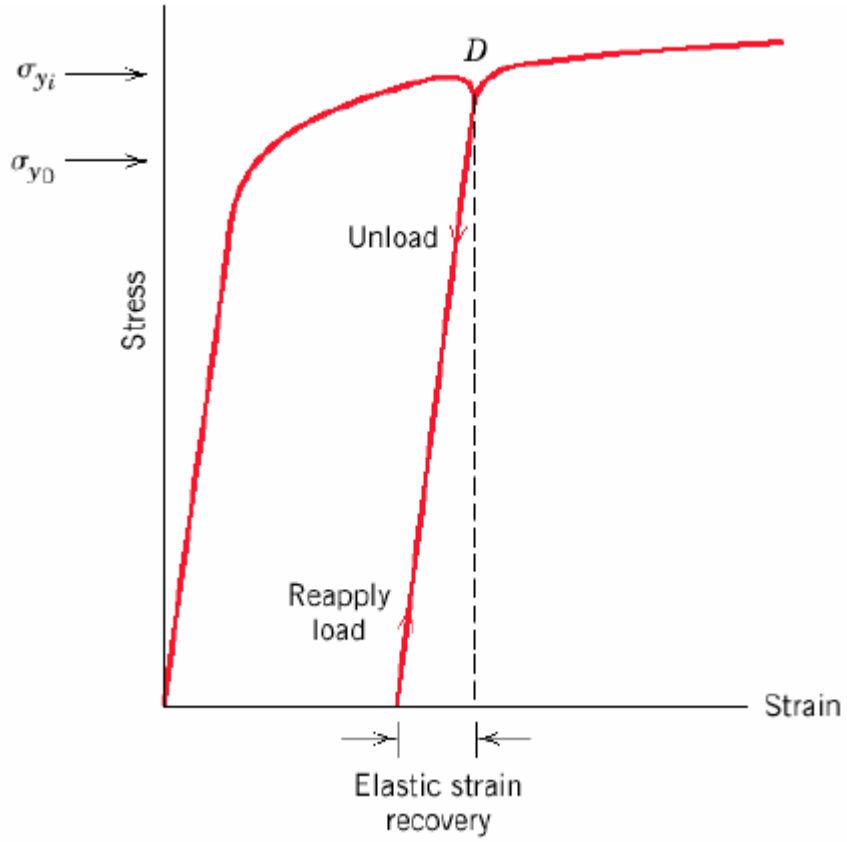
d) To compute the change in length, Δl, it is first necessary to determine the strain that is produced by a stress of 345 MPa. This is accomplished by locating the stress point on the stress–strain curve, point A, and reading the corresponding strain from the strain axis, which is approximately 0.06. Inasmuch as $l_o = 250mm$, we have

$$\Delta l = \varepsilon l_o = (0.06)(250mm) = 15\,mm$$

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Elastic Recovery During Plastic Deformation

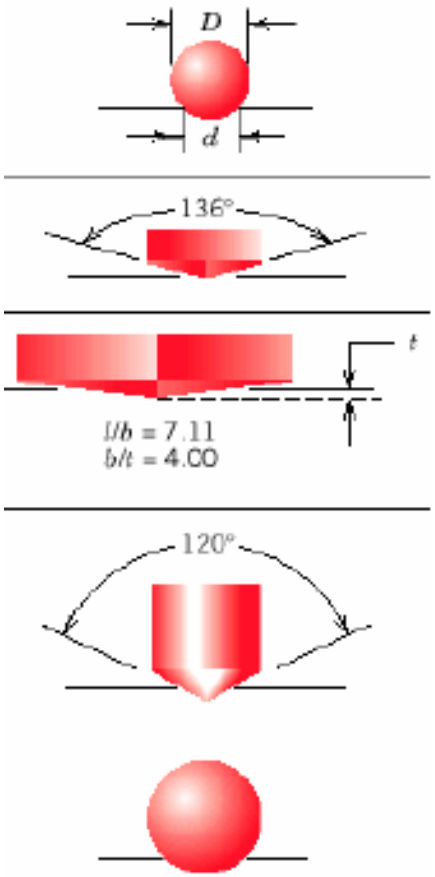
- If a material is deformed plastically and the stress is then released, the material ends up with a permanent strain.
- If the stress is reapplied, the material again responds elastically at the beginning up to a new yield point that is higher than the original yield point.
- The amount of elastic strain that it will take before reaching the yield point is called **elastic strain recovery**.



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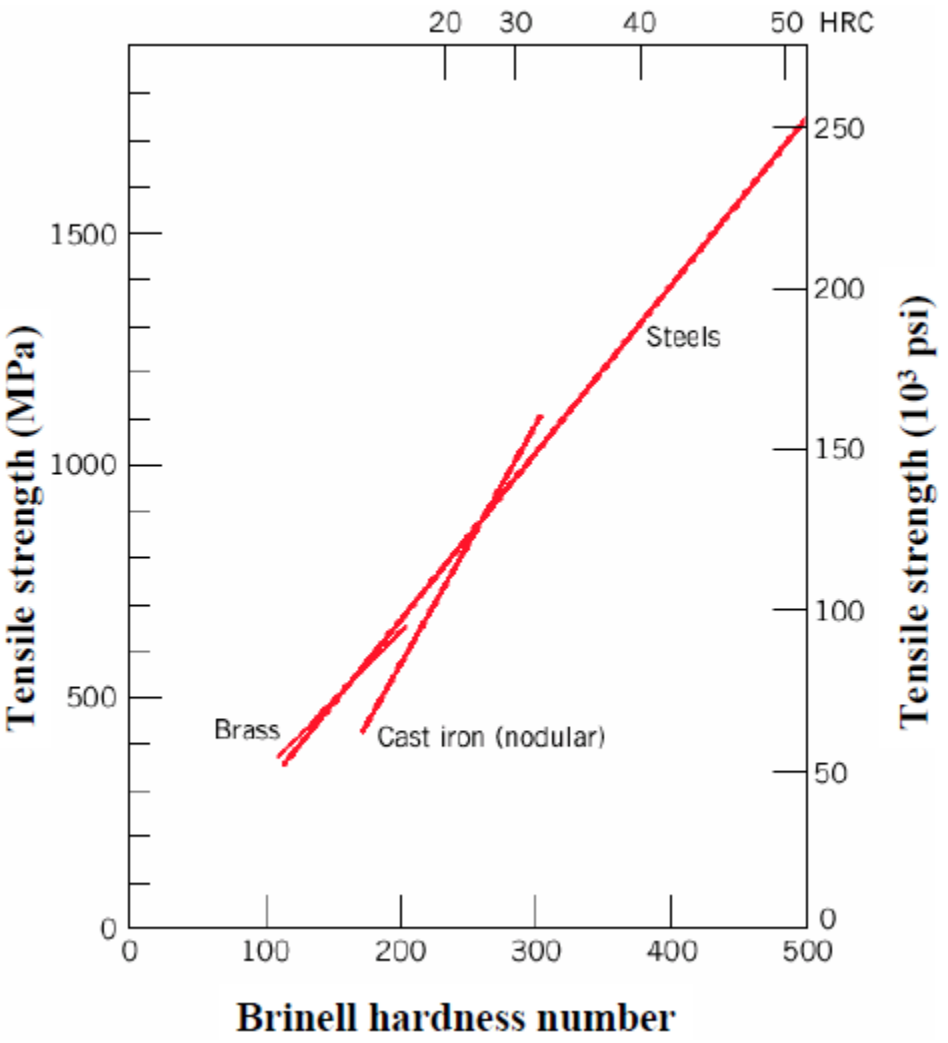
Hardness (I)

- **Hardness is a measure of the material's resistance to localized plastic deformation** (e.g. dent or scratch)
- A qualitative Moh's scale, is determined by the ability of a material to scratch another material: from 1 (softest = talc) to 10 (hardest = diamond).
- Different types of quantitative hardness test has been designed (Rockwell, Brinell, Vickers, etc.).
- Usually a small indenter (sphere, cone, or pyramid) is forced into the surface of a material under conditions of controlled magnitude and rate of loading. The depth or size of indentation is measured.
- The tests somewhat approximate, but are popular because they are easy and non-destructive (except for the small dent).



Hardness (II)

- Both tensile strength and hardness may be regarded as degree of resistance to plastic deformation.
- **Hardness is proportional to the tensile strength** – but note that the proportionality constant is different for different materials.



What are the limits of "safe" deformation?

- **Design stress:** σ_d = N' σ_c

  where σ_c = maximum anticipated stress,

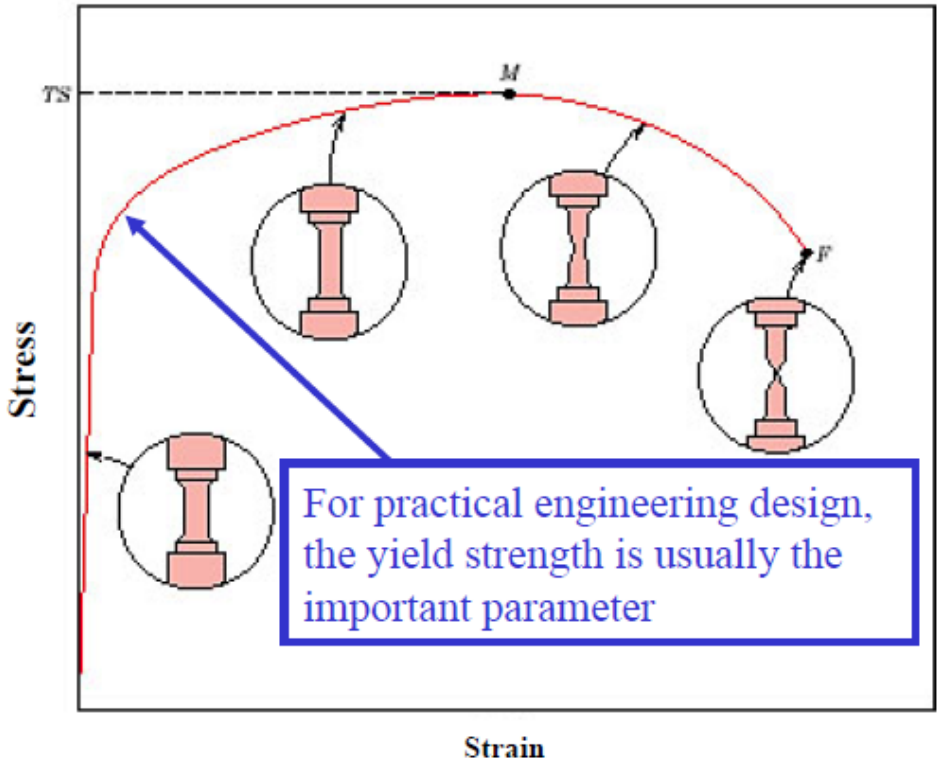
  N' is the "design factor" > 1.

  Want to make sure that $\sigma_d < \sigma_y$

- **Safe or working stress**

  $\sigma_w = \sigma_y/\mathrm{N}$

  where N is "factor of safety" > 1.



Thank You